

Omar Tawfik

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PROFESSIONAL SUMMARY

Data Scientist with 4+ years building and deploying predictive models, causal inference frameworks, and forecasting systems for high-stakes decisions. Skilled in Python, SQL, statistical modeling, feature engineering, and ETL pipelines, with applied AI and LLM engineering experience across production deployments.

EXPERIENCE

Data Scientist

2026 – Present

Office of Labor Relations & Collective Bargaining (OLRCB)

- Quantified \$31M in potential settlement liability for a 26,000+ person workforce by building a compensation and back-pay prediction pipeline on payroll and demographic data, applying feature engineering, regression, and cross-validated statistical modeling.
- Replaced single-point fiscal estimates with probabilistic exposure ranges (P5-P95) by engineering multi-year forecasting models across 5 negotiation strategies, combining time-series forecasting and stochastic sensitivity analysis.
- Reduced projected fiscal exposure by approximately \$9M by developing compensation models, labor market analyses, and costing scenarios that informed bargaining strategy across 10+ union negotiations impacting 26,000+ employees.
- Architected a centralized analytics and compliance framework with SQL-based data models and MLOps practices, eliminating manual tracking gaps across 20+ compliance obligations.
- Built an end-to-end causal inference platform (Python, Flask, SQL) combining DiD, Synthetic Control, and PSM via inverse-variance weighting, validated with Monte Carlo simulations.

Data Scientist

Jul 2023 – Nov 2024

Booz Allen Hamilton Inc.

- Architected an end-to-end ETL pipeline using SQL and Azure that replaced manual reporting workflows, lifting team productivity 10x and improving data accuracy 50% across 10+ biweekly executive reports.
- Led a 6-person team standardizing KPI definitions across 4 business units and automating reporting with Python and Spark, cutting manual reporting effort by ~10 hours per week and delivering consistent Tableau dashboards to executives.
- Redesigned Salesforce LTV tracking infrastructure, improving response time by 35% for 5,000+ monthly active users.

PROJECTS

DocMind: RAG Document Intelligence Pipeline

Jan 2026 – Mar 2026

<https://docmind-rag-pipeline.onrender.com/>

Independent

- Built and deployed a full RAG pipeline (Python, Flask, Render) ingesting PDF, TXT, and Markdown documents, answering user questions grounded in source material with citations and relevance scores.
- Integrated Groq-hosted Llama 3.3 70B with fastembed (ONNX) embeddings and a Qdrant vector store; validated retrieval with a 10-question harness, achieving 100% hit-rate@6 and 0.95 MRR.

ReviewGuard: Fraudulent Review Detection Engine

May 2024 – Aug 2024

Independent

- Engineered 18 behavioral and linguistic features with TF-IDF vectorization (10,000 features) to classify fraudulent reviews across 42,000+ records.
- Benchmarked Logistic Regression, Random Forest, Linear SVM, and Gradient Boosting using stratified 5-fold cross-validation, achieving 90.68% accuracy and 90.62 F1-score.

SKILLS

Machine Learning & Statistics: Predictive Modeling, Feature Engineering, Causal Inference (DiD, Synthetic Control, PSM), Time Series Forecasting, NLP & Text Classification, Statistical Modeling, Regression, Classification, Experimental Design

AI & LLM Engineering: RAG, LLM APIs (Groq/Llama), Vector Databases (Qdrant), Prompt Engineering, Retrieval Evaluation, GenAI

Programming & Tools: Python, SQL, scikit-learn, Flask, Git/GitHub, MLOps

Data Engineering & ETL: ETL Pipeline Development, Apache Spark, Databricks, Snowflake, Data Cleansing

Cloud & Visualization: Azure, Tableau, Power BI, Qlik, Chart.js

EDUCATION

University of Maryland, Smith School of Business

Master of Science, Data Science. GPA: 4.0.

May 2026

University of Maryland

Bachelor of Science, Operations Management & Business Analytics. GPA: 3.9.

May 2023